

pline-vec

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Contents

1	Introduction	1
2	Examples	1
2.1	changing the main orientation, and sub orientation within it . . .	1
2.2	parametric lines	1
2.3	Parametric surfaces	1
2.4	parametric solids	2
2.5	drawing clipped planes	2
2.6	intersecting planes	3
2.7	math function documentation	3

1 Introduction

I really enjoy making 3D artwork using *TikZ*, especially animations. I really wanted to z-buffer multiple surfaces at once and graph intersecting planes, but these tasks proved to be very difficult using the tools available, namely *pgfplots* and *tikz-3dplot*.

The goal of the package *pline-vec* is to overcome these obstacles. I have already implemented a rigorous z-buffering algorithm that can handle multiple surfaces, and intersecting planes has been made possible, though it lacks automatic z-sorting at this point. That will be fixed.

An additional feature of *pline-vec* is that it defines many useful math functions and operations. For example, there are lua-based *TikZ* functions for rotating objects in 3D, traversing spheres, taking dot products and cross product components, as well as for easily setting multiple *pgfmath* macros at once.

2 Examples

2.1 changing the main orientation, and sub orientation within it

2.2 parametric lines

We will z-buffer lines, surfaces, and solids all at once.

2.3 Parametric surfaces

The code to generate non-intersecting (and also non-tangent) surfaces is complete. The user specifies the domain, range, samples, and **Lua** math function, and pline-vec draws the surface for them. In the future, I will implement the ability to add more keys for things like opacity and color for individual surfaces.

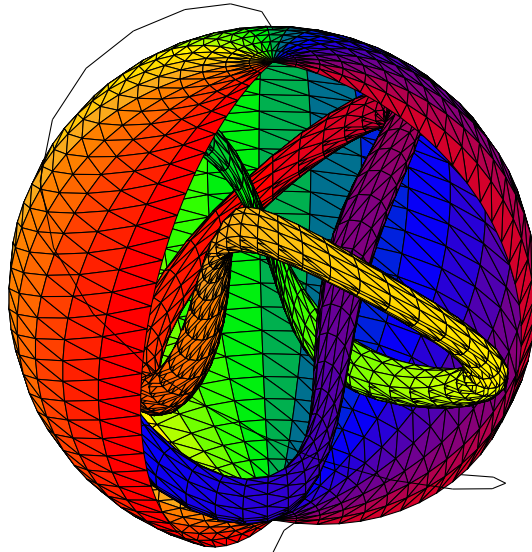


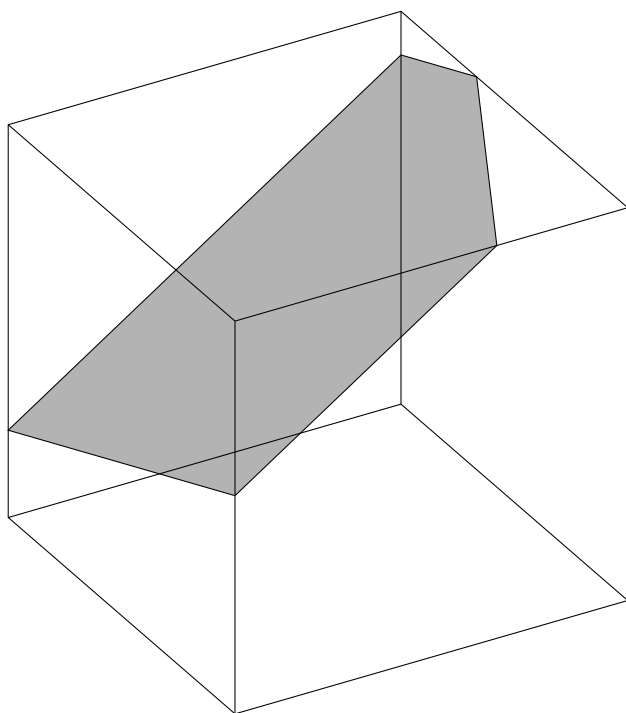
Figure 1:
A parametric surface

2.4 parametric solids

I intend to implement code for rendering parametric solids, since the z-buffering technique I use will work for them too.

2.5 drawing clipped planes

pline-vec lets you draw clipped planes. In the future, this will be done similarly to how it is done for parametric surfaces. For example;



2.6 intersecting planes

2.7 math function documentation